

Normalizing flows

→ Basically, it is about using the fact that

$$f_Y(y)/dy = f_X(x)/dx$$

where $y=g(x)$ to transform a simple distribution into a much more complex one.

$$\text{therefore } f_Y(g(x)) = f_X(x) \left| \frac{dx}{dy} \right|$$

$$f_Y(g(x)) = f_X(x) \left| \frac{dy}{dx} \right|^{-1} = f_X(x) \left| J_g^{-1} \right|$$

In a more general way (multivariate distributions):

$$\mathbf{Z} \sim \pi(\mathbf{Z}), \mathbf{X} = \mathbf{g}(\mathbf{Z}), \mathbf{Z} = \mathbf{g}^{-1}(\mathbf{X})$$

$$p(\mathbf{X}) = \pi(\mathbf{Z}) \left| \det \frac{d\mathbf{Z}}{d\mathbf{X}} \right| = \pi(\mathbf{Z}) \left| \det \frac{d\mathbf{g}^{-1}(\mathbf{X})}{d\mathbf{X}} \right|$$

$$= \pi(\mathbf{Z}) \left| \det \frac{d\mathbf{g}(\mathbf{Z})}{d\mathbf{Z}} \right|^{-1}$$

→ Basics

$\vec{X}$ a D -dimensional real vector. We would like to define a joint distribution over $\vec{X}$.

The idea is to express $\vec{X}$ as a transformation of a latent variable $\vec{u}$, which follows the prior distribution $p_u(\vec{u})$.

Then

$$\vec{X} = T(\vec{u}) \text{ where } \vec{u} \sim p_u(\vec{u})$$

→ Change of variable theorem

In order to express the joint distribution of $\vec{X}$, we need to express $p_X(\vec{X})$ in terms of $p_u(\vec{u})$ using a change of variables.

$$\int dx p(x) = \int dz \pi(z) = 1$$

$$p(x) = \pi(z) \left| \frac{dz}{dx} \right| = \pi(g^{-1}(x)) \left| \frac{dg^{-1}}{dx} \right|$$

Using the previous notation

$$p_X(\vec{X}) = p_u(T^{-1}(\mathbf{u})) \left| \frac{dT^{-1}}{d\mathbf{u}} \right|$$

→ Kullback-Leibler divergence

→ Measure of how one prob. dist. is different from a second.

$$DKL(P||Q) = \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right)$$

$$= \sum_{x \in \mathcal{X}} P(x) [\log(P(x)) - \log(Q(x))]$$

that is, this is the difference of log distributions weighted by $P(x)$.

Neyman-Pearson lemma

It shows that the most powerful way to distinguish between the two distributions P and Q based on an observation Y is through the log of the ratio of their likelihoods $\log P(Y) - \log Q(Y)$.

$$p_X(\vec{x}) d\vec{x} = p_u(\vec{u}) d\vec{u} \quad \vec{x} = T(\vec{u})$$

$$p_X(\vec{x}) = p_u(\vec{u}) \left| \det J_{T^{-1}(\vec{x})} \right| \quad \frac{dT^{-1}(\vec{x})}{d\vec{x}}$$

$$p_X(\vec{x}) = p_u(T^{-1}(\vec{x})) \left| \det J_{T^{-1}(\vec{x})} \right|$$

$$\vec{u}_{i-1} \sim p_{i-1}(\vec{u}_{i-1})$$

$$\vec{u}_i = T_i(\vec{u}_{i-1})$$

$$p_i(u_i) = p_{i-1}(T_i^{-1}(u_i)) \left| \det J_{T_i^{-1}} \right|$$

Neural Ordinary Differential Equations

Basically, the normalizing flows are based on the idea that we are going to discretely transform a function as

$$h_{t+1} = h_t + f(h_t, \theta_t)$$

$$t \in \{0 \dots T\} \text{ and } h_t \in \mathbb{R}^D$$

In the continuous limit

$$\frac{dh(t)}{dt} = f(h(t), t, \theta)$$

$$dh(t) = f dt$$

$$\int_0^T h(t) dt = \int_0^T f dt$$

$$h(t) - h(0) = \int_0^t f dt$$

Theorem 1 (Instantaneous change of variables). $Z(t)$ finite continuous random variable with prob. $p(Z(t))$.

$\frac{dz}{dt} = f(z(t), t)$ f is uniformly Lipschitz continuous in z and t . Then the change in log prob. follows

$$\frac{\partial \log p(z(t))}{\partial t} = -\text{tr} \left(\frac{df}{dz(t)} \right)$$

$$\log p(z(t_1)) = \log p(z(t_0)) - \int_{t_0}^{t_1} \text{Tr} \left(\frac{df}{dz(t)} \right) dt$$

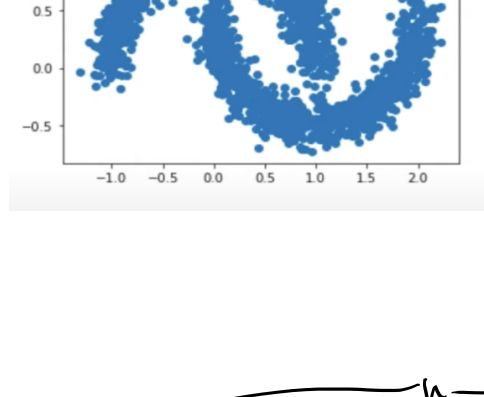
* Hutchinson's trace:

Given any matrix $W \in \mathbb{R}^{n \times n}$, and any random $u \in \mathbb{R}$, such as $\mathbb{E}[uu^T] = \mathbb{I}$, then $\mathbb{E}[u^T W u] = \text{Tr}(W)$

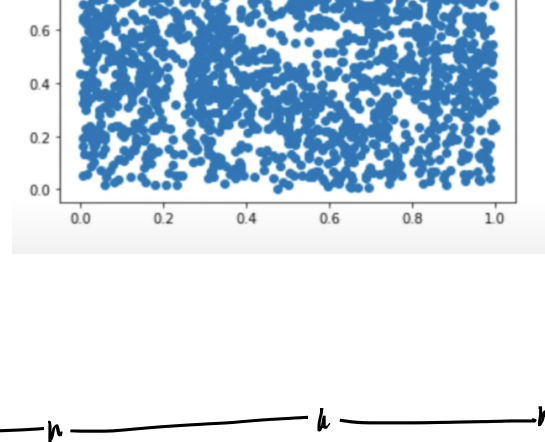
Normalizing flows

→ Motivation

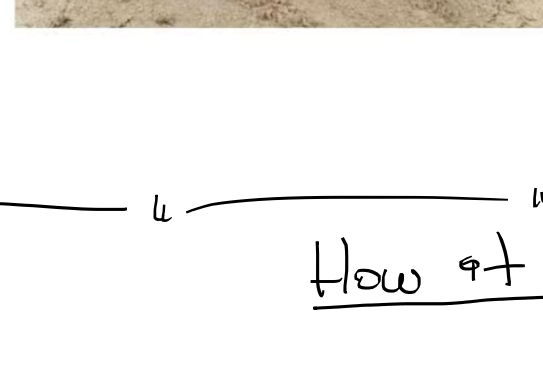
Given a set of random variables with unknown PDF. Learn how to "destroy" it



f
← f^{-1}



How to construct a sand castle?



destroy →



← reverse the process

How it works?

$\vec{x} \in \mathbb{R}^D$ and define

$\vec{u} = T^{-1}(\vec{x})$, where $\vec{u} \sim p_u(\vec{u})$ (known)

and $T^{-1}: \mathbb{R}^D \rightarrow \mathbb{R}^D$ is a Diffeomorphism

The target distribution $p_x(\vec{x})$ is unknown, but we have the data $\mathcal{D} = \{\vec{x}_i\}_{i=1}^M$, given by the simulator.

Then, we can "flow" from p_u to p_x , by using

$$p_x(\vec{x}) = p_u(\vec{u}) |\det J_T(\vec{u})|^{-1}$$

$$= p_u(T^{-1}(\vec{x})) |\det J_{T^{-1}}(\vec{x})|. \quad (1)$$

$$[J_T(\vec{u})]_{ij} \equiv \frac{\partial T_i}{\partial u_j}, \quad T_i \equiv T(u_i)$$

However, it's hard to define a bijective T that is able to "wrap" a simple distribution to any target distribution.

Solution: Given T_1 and T_2 (diffeomorphisms),

$T_2 \circ T_1 \equiv T_2(T_1)$ is also a diffeomorphism!

and

$$\det J_{T_2 \circ T_1}(\vec{u}) = \det J_{T_2} \cdot \det J_{T_1}$$

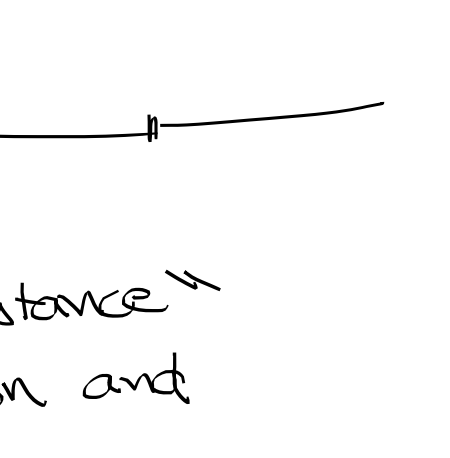
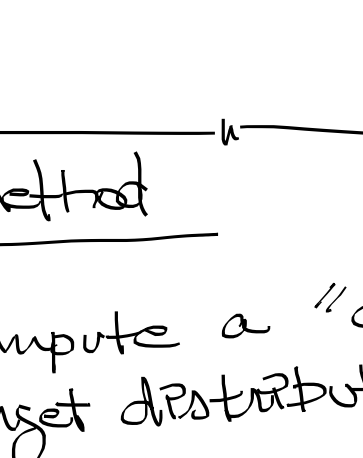
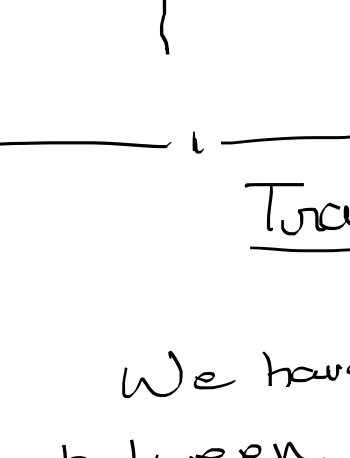
Therefore

$$\log p_x(\vec{x}) = \log p_K(\vec{u}_K) = \log p_0(\vec{u}_0) + \sum_{i=1}^K |\det J_{T_i}(\vec{u}_{i-1})|$$

$$= \log p_0(T_0^{-1}(\vec{x})) - \sum_{i=1}^K |\det J_{T_i}(\vec{x})|,$$

where we have used $\det \frac{df^{-1}}{dx} = \det \left(\frac{df}{du} \right)^{-1}$
 $f^{-1}(x) = u$, $f(u) = x$ and $\det(A^{-1}) = \det(A)^{-1}$.

$$\textcircled{u_0} \xrightarrow{f_1(u_0)} \textcircled{u_1} \dots \xrightarrow{f_i(u_{i-1})} \textcircled{u_i} \dots \xrightarrow{f_K(u_{K-1})} \textcircled{u_K} \equiv x$$



Training method

We have to compute a "distance" between the target distribution and the one given by the flow.

The "distance" between two distributions is given by the Kullback-Leibler (KL) divergence

$$D_{KL}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right)$$

Imagine that $T_\theta(u)$ is parametrized by $\vec{\theta}$,

then

$$D_{KL}[P_x^*(\vec{x}) \parallel P_x(\vec{x}; \vec{\theta})] =$$

$$- \mathbb{E}_{p^*(\vec{x})} [\log P_x(\vec{x}; \vec{\theta})] + \text{const.}$$

$$= - \mathbb{E}_{p^*(x)} [\log p_u(T_\theta^{-1}(\vec{x})) + \log |\det J_{T_\theta}(\vec{x})|] + \text{const.}$$

Assuming that we have $\{\vec{x}_n\}_{n=1}^N$ from $p_x^*(\vec{x})$ and using the approximation $\mathbb{E}_{f(x)}[g(x)]$

$$\mathbb{E}_{f(x)}[g(x)] \approx \frac{1}{N} \sum_{i=1}^N g(x_i), \quad x_i \sim f(x_i)$$

$$\therefore \mathcal{L}(\theta) \equiv D_{KL}[P_x^*(\vec{x}) \parallel P_x(\vec{x}; \vec{\theta})] \approx - \frac{1}{N} \sum_{n=1}^N [\log p_u(T_\theta^{-1}(x_n)) + \log |\det J_{T_\theta}(x_n)|] + \text{const.}$$

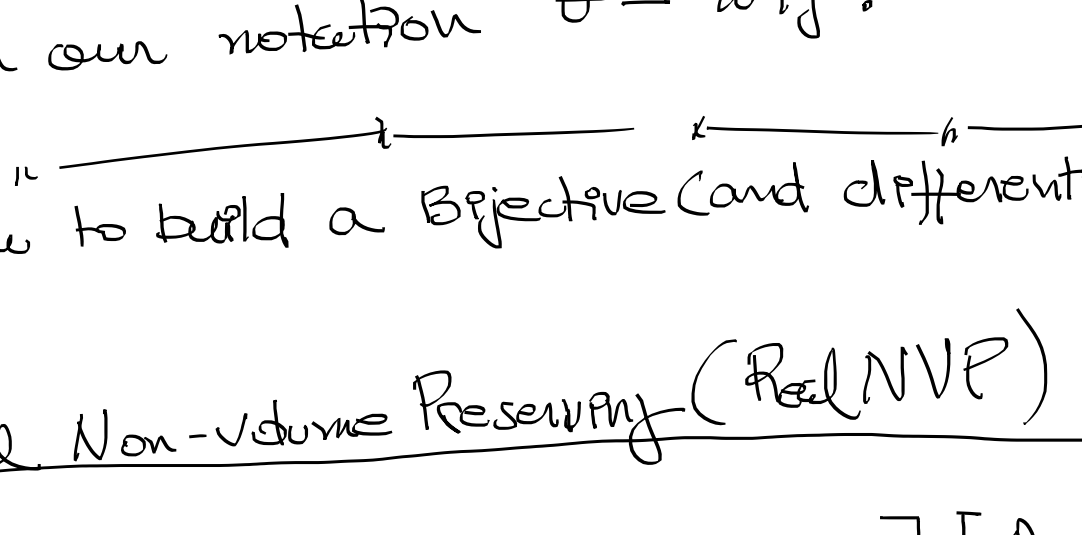
Therefore, the training procedure is to minimize $\mathcal{L}(\theta)$ through $\vec{\nabla} \mathcal{L}$.

But what is $T_\theta(\vec{x})$??

x input; $y = T(x)$ output; L number of layers; $W^l \equiv (w_{jk}^l)$ the weights between the k -th node in layer $l-1$ and the j -th node in the layer l .

$$T(x) := f^L(W^L f^{L-1}(W^{L-1} \dots f^1(W^1 x) \dots))$$

f is the activation function, or "perception". For instance ReLU (rectified linear unit)



In our notation $\theta \equiv w_{ij}^l$.

How to build a Bijective (and differentiable) XIN?

Real Non-Volume Preserving (RealNVP)

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = T_\theta \left(\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} \right) = \begin{bmatrix} u_1 \\ \rho \phi(u_1) u_2 \end{bmatrix} + \begin{bmatrix} 0 \\ m_\phi(u_1) \end{bmatrix}$$

The inversion is trivially

$$u_1 = x_1; \quad u_2 = e^{-\phi(u_1)} (x_2 - m_\phi(x_1))$$

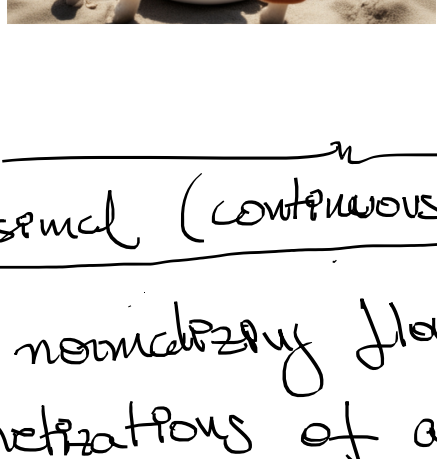
and the Jacobian is trivially

$$\log J_{T_\theta} = \log \left(\prod_{i=1}^n e^{\phi(u_i)} \right) = \sum_{i=1}^n \phi(u_i).$$

If the training set is a set of images of cats, beaches and mushrooms,



and...



Infinitesimal (continuous) flows

The discrete normalizing flows can be viewed as discretizations of a first order ODE:

$$\frac{d\vec{x}(t)}{dt} = F(\vec{x}(t); \theta(t)), \quad \vec{x}(t=0) = \vec{u}$$

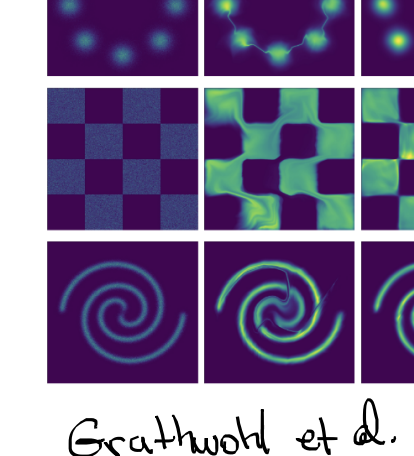
The discrete version (Euler's method):

$$x_{n+1} = x_n + \epsilon F(x_n; \theta_n)$$

The change of variables formula is

$$\frac{d}{dt} \log(p(\vec{x}(t))) = -\text{Tr} \left(\frac{dF(\vec{x}(t))}{d\vec{x}(t)} \right).$$

We no longer need to compute the determinant and F can be anything.



Grathwohl et al. (2018)